CYCLIC PARAVECTOR OPERATORS AND THE LAGUERRE–CLIFFORD TOWER: CLOSED COMPUTATIONS TOWARD A MATRIX FORMULATION OF THE RIEMANN HYPOTHESIS (LAYERS I AND II), WITH LEAN 4 VERIFICATION

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Abstract. This report collects the computations that have been closed in a programme whose aim is to restate the Riemann hypothesis (RH) as a statement about matrices and to examine whether the paravector nullity theory of cyclic Clifford operators can serve as a tool for it. Layer I is algebraic: it upgrades the nullity formula for cyclic paravector operators to a quantitative theory (exact singular values in every signature, two-sided near-kernel bounds, positivity of block circulant and block Toeplitz matrices through the Lorentz cone, a twisted-circulant zero count, and a chiral (Dirac-type) structure of persymmetric operators). Layer II is analytic: it writes the Laguerre forms of Riemann's Ξ as norms of paravectors in the split Clifford algebras $\mathrm{Cl}(n,n)$, proves single-point matrix criteria equivalent to RH, a depth bound $O(\log\gamma)$ for the level at which an isolated off-line zero is detected, and an index theorem: the number of negative eigenvalues of a single-point Hankel matrix counts the off-line zeros. *Nothing here proves RH*; every equivalence moves the difficulty rather than removing it. Each item carries a status tag, and the finite algebraic cores are formalised in Lean 4 in the repository `Cyclic-Fueter-Theory-Lean000`. Novelty is not claimed: several items are classical and are cited as such, and for others the literature check is recorded as preliminary.

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1. Status conventions and scope

Every statement carries one tag.

[Closed]: proved in this report, with every step written out.
[Closed-cond]: proved here assuming the cited classical results listed in §3; no other assumption.
[Classical]: a known theorem, stated for completeness with a reference; our proof, if given, is only a convenience.
[Numerical]: observed in exact or high-precision computation; not a theorem.
[Conjecture]: a precise statement that is supported numerically but not proved.

A Lean tag `[Lean: name]` means the displayed core is a theorem of the Lean project, built by the GitHub Actions workflow of the repository with no `sorry`, `admit` or project axiom. "Lean core" means that the algebraic heart is formalised and the remaining steps (spectral multiplicities, analytic limits, citations) are not. Numerical checks refer to the Python scripts in `verification/` (NumPy, exact rationals via `fractions`, and 110-digit decimal arithmetic).

What the report does and does not do. It records closed computations. It does not prove RH, does not claim that the tools below reach RH, and assesses novelty only provisionally (§8).

2. Conventions

2.1. **Clifford algebras and paravectors.** $\mathrm{Cl}(p,q)$ is the real Clifford algebra with generators $e_1,\dots,e_n$, $n=p+q$, $e_a^2=s_a$ with $s_a=+1$ for $a\le p$ and $s_a=-1$ for $a>p$, and $e_ae_b=-e_be_a$ for $a\neq b$. A *paravector* is $p=p_0+\sum_a p_ae_a$ with real or complex coefficients. Its Clifford norm and Hermitian norm are

$$N(p)=p\bar p=p_0^2-\sum_a s_ap_a^2,\qquad \|p\|^2=\sum_{j=0}^{n}|p_j|^2.$$

$L(p)$ denotes left multiplication on the 2^n-dimensional algebra, written in the blade basis. In the Lean project, Mathlib's convention $\iota(v)^2=Q(v)$ is used; $\mathrm{Cl}(n,0)$ corresponds to $Q(v)=\sum v_i^2$ and the manuscript convention $\mathrm{Cl}(0,n)$ of the existing core to $Q(v)=-\sum v_i^2$.

2.2. **Riemann's Ξ and the Laguerre forms.** $\Xi(t)=\xi(\frac12+it)$ is real, even and entire of order one. Its zeros are written t_ρ (so $\rho=\frac12+it_\rho$); RH says that all t_ρ are real. For a real function f and a real point x we use the Taylor coefficients $c_k=f^{(k)}(x)/k!$ and the Laguerre forms

$$L_n[f](x)=\sum_{k=0}^{2n}(-1)^{n+k}c_kc_{2n-k}=\sum_{k=0}^{2n}\frac{(-1)^{n+k}}{k!(2n-k)!}f^{(k)}(x)f^{(2n-k)}(x). \tag{1}$$

For a real point x that is not a zero, the *atoms seen from* x are $w_\rho(x) = (x - t_\rho)^{-2}$ (one for each zero, with multiplicity), with power sums $p_k(x) = \sum_\rho w_\rho(x)^k$ and elementary symmetric functions $e_n(x) = e_n(\{w_\rho(x)\})$.

3. Cited classical results

(C1) $\Xi(t) = \Xi(0)\prod_\rho(1 - t^2/t_\rho^2)$ over the zeros with $\operatorname{Re} t_\rho > 0$ (or one of each pair $\pm t_\rho$), and $\sum_\rho |t_\rho|^{-2} < \infty$ [17].
(C2) Every zero has $|\operatorname{Re} t_\rho| > 14$ [17].
(C3) $N(T+1) - N(T) = O(\log T)$ (Riemann–von Mangoldt) [17].
(C4) Hamburger's theorem: a real sequence whose Hankel matrices are all positive semidefinite is a moment sequence of a positive measure on $\mathbb{R}$; the measure is unique when the moments grow at most geometrically.
(C5) Edrei's theorem (Aissen–Schoenberg–Whitney–Edrei): $(1, c_1, c_2, \dots)$ is totally nonnegative iff $\sum c_k z^k = e^{\gamma z}\prod(1 + \beta_i z)/\prod(1 - \alpha_i z)$ with $\alpha_i, \beta_i, \gamma \geq 0$, $\sum \alpha_i + \sum \beta_i < \infty$ [1, 7, 18].
(C6) Li's criterion: RH iff $\lambda_n = \sum_\rho[1 - (1 - 1/\rho)^n] \geq 0$ for all $n \geq 1$ [12, 2].
(C7) Darroch's theorem: the mode M of a sum of independent Bernoulli variables with mean μ satisfies $M \leq \lfloor \mu \rfloor + 1$ [6, 14].
(C8) Newton's inequalities $e_n^2 \geq (1 + \frac{1}{n})(1 + \frac{1}{N-n})e_{n-1}e_{n+1}$ for N real numbers [9].
(C9) Weyl's inequality $|\lambda_k(A + E) - \lambda_k(A)| \leq \|E\|$ for Hermitian A, E, and Cauchy interlacing.
(C10) Jensen's identity $|f(x + iy)|^2 = \sum_n L_n[f](x)y^{2n}$ and the criterion "RH iff $L_n[\Xi](x) \geq 0$ for all n and real x" (Jensen; Patrick; Csordas–Varga) [5, 4].

4. Layer I: a quantitative paravector nullity theory

4.1. **The nullity formula (I.1).** The kernel formula for cyclic paravector operators, $\dim\ker = 2^n N_0 + 2^{n-1}N_{\mathrm{iso}}$ (zero modes and nonzero isotropic modes), is the starting point. It is formalised in the existing core of the repository for complex coefficients and for complexified real data `[Lean: sum_paravectorKernel_finrank_eq_corrected_formula]`.

4.2. **Spectrum in** $\mathrm{Cl}(n, 0)$ **(I.2).**

Lemma 4.1. [Closed] *In* $\mathrm{Cl}(n, 0)$, $L(p)$ *is symmetric with eigenvalues* $p_0 \pm |\vec{p}|$, *each of multiplicity* 2^{n-1}.

Proof. $E_a = L(e_a)$ permutes the blade basis up to sign, so it is orthogonal; $E_a^2 = I$ gives $E_a^{\mathsf{T}} = E_a$. For $a \neq b$ the E_a anticommute, hence $(L(p) - p_0)^2 = (\sum p_a E_a)^2 = |\vec{p}|^2 I$. Each E_a maps the blade e_A to $\pm e_{A \triangle \{a\}} \neq e_A$, so $\operatorname{tr} E_a = 0$ and the eigenvalues $p_0 \pm |\vec{p}|$ occur equally often. □

Lean core: in the Clifford algebra of *any* quadratic form, $(L_p - r)^2 = Q(v)$ for $p = r + \iota v$, every eigenvalue satisfies $(\mu - r)^2 = Q(v)$, in $\mathrm{Cl}(n, 0)$ every eigenvalue is $r \pm |v|$, and $r \geq |v|$ forces all eigenvalues to be non-negative `[Lean: paravectorMul_quadratic_relation, paravectorMul_eigenvalue, euclidean_paravector_eigenvalue, euclidean_paravector_eigenvalue_nonneg]`. Multiplicities are not formalised.

4.3. **Singular values in every signature (I.3, I.4).**

Theorem 4.2. [Closed] *For a complex paravector* p *in* $\mathrm{Cl}(p, q)$ *the singular values of* $L(p)$ *are*

$$\sigma_\pm^2 = \|p\|^2 \pm \sqrt{\|p\|^4 - |N(p)|^2},$$

each with multiplicity 2^{n-1}. *In particular* $\sigma_+\sigma_- = |N(p)|$ *and* $\sigma_+^2 + \sigma_-^2 = 2\|p\|^2$.

Proof. (i) $E_a^{\mathsf{T}} = s_a E_a$ as in Lemma 4.1. (ii) With $\alpha_a = s_a \bar{p}_a$, $\beta_a = p_a$ one has $L(p)^* = L(\bar{p}_0 + \alpha)$, so $L(p)^* L(p) = L((\bar{p}_0 + \alpha)(p_0 + \beta))$. Its scalar part is $|p_0|^2 + \sum s_a \alpha_a \beta_a = \|p\|^2$, and the rest is $L(X)$ with $X = v + B$, $v = \bar{p}_0 \beta + p_0 \alpha$, $B = \alpha \wedge \beta$. (iii) v and B lie in the Clifford algebra of the plane $\mathrm{span}(\alpha, \beta)$; there the bivector anticommutes with vectors and squares to a scalar (Lemma 4.3), so $X^2 = v^2 + B^2$ is a scalar c. Since X is Hermitian and traceless, its eigenvalues are $\pm\sqrt{c}$ with equal multiplicity. (iv) $L(p)L(\bar{p}) = N(p)$ and the grade involution conjugates $L(p)$ to $L(\bar{p})$, so $\lambda_+ \lambda_- = |N(p)|^2$ for the two eigenvalues of $L(p)^* L(p)$; with $\lambda_+ + \lambda_- = 2\|p\|^2$ this gives the formula. □

Lemma 4.3. [Closed] *In any algebra, if $u^2 = s$, $w^2 = t$ and $uw + wu = p$ are scalars, then $e = uw - wu$ satisfies $eu + ue = 0$, $ew + we = 0$ and $e^2 = p^2 - 4st$. In a Clifford algebra this holds for $u = \iota v$, $w = \iota v'$ with $p = \mathrm{polar}(v, v')$.* `[Lean: commutator_of_scalar_relations, clifford_commutator_relations]`

Proof. $eu + ue = u^2 w - w u^2$, $ew + we = uw^2 - w^2 u$ and $e^2 = (uw + wu)^2 - 2(uw^2 u + wu^2 w)$ are identities in the free ring; substitute. □

Corollary 4.4. [Closed] *For the persymmetric cyclic operator $H_N(a)$ with symbol P and $T = N(P)$,*

$$\min_k \frac{|T(\omega_k)|}{\sqrt{2}\,\|P(\omega_k)\|} \;\le\; \sigma_{\min}(H_N(a)) \;\le\; \min_k \frac{|T(\omega_k)|}{\|P(\omega_k)\|},$$

and both bounds are attained.

Proof. $H_N(a)$ is unitarily equivalent to $\bigoplus_k L(P(\omega_k))$ composed with a mode permutation, and $\sigma_-^2 = |N|^2/(\|p\|^2 + \sqrt{\|p\|^4 - |N|^2}) \in [|N|^2/(2\|p\|^2), |N|^2/\|p\|^2]$. Attainment: $N = 0$ resp. $\|p\|^4 = |N|^2$. □

Numerical: each step (i)–(iv) checked in 16 signatures; the bound ratios reach 1.000 (`new1_svd_proof.py`, `test_A.py`).

4.4. **Block circulant positivity (I.5).**

Theorem 4.5. [Closed] *Let C be the block circulant $C_{rc} = L(a_{(c-r) \bmod N})$ in $\mathrm{Cl}(n, 0)$ with $a_{-d} = a_d$ real. Then $\lambda_{\min}(C) = \min_k [P_0 - |\vec{P}|](\omega_k)$, $C \succeq 0$ iff every $P(\omega_k)$ lies in the closed Lorentz cone $\{P_0 \ge |\vec{P}|\}$, and* $\dim\ker C = 2^{n-1} \#\{k : \text{boundary}, P \neq 0\} + 2^n \#\{k : P = 0\}$.

Proof. The discrete Fourier transform block-diagonalises C into $L(P(\omega_k))$; apply Lemma 4.1. □

The Fourier reduction is formalised in the core for the $(r + c)$ convention `[Lean: FourierReduction]`; the cone direction of the eigenvalue statement is `[Lean: euclidean_paravector_eigenvalue_nonneg]`.

4.5. **Block Toeplitz matrices: finite-size bounds (I.6).**

Let $F(\theta) = L(P(\theta))$ with $P(\theta) = b_0 + 2\,\mathrm{Re}\sum_{d \ge 1} b_d e^{id\theta}$, $T_M = [\hat{F}_{r-c}]$, $\lambda(\theta) = P_0 - |\vec{P}|$, $m = \min \lambda$, and $s_M = 4\sin^2 \frac{\pi}{2(M+1)}$, the least eigenvalue of $\mathrm{tridiag}(-1, 2, -1)$.

Theorem 4.6. [Closed] *Let θ_0 be a minimiser and u a unit minimal eigenvector of $F(\theta_0)$.*

- (a) $m \le \lambda_{\min}(T_M) \le m + K s_M$ *with* $K = \sup_\theta \frac{u^* F(\theta) u - m}{2 - 2\cos(\theta - \theta_0)}$.
- (b) *If θ_0 is the only minimiser,* $\lambda_{\min}(T_M) \ge m + c\, s_M$ *with* $c = \inf_\theta \frac{\lambda(\theta) - m}{2 - 2\cos(\theta - \theta_0)}$.
- (c) *For even real symbols with the two minimisers $\pm\theta_0$,* $\lambda_{\min}(T_M) \ge m + c_2 \mu_M$ *with* $c_2 = \inf \frac{\lambda - m}{4(\cos\theta - \cos\theta_0)^2}$ *and* $\mu_M \ge s_M s_{M+1}$.
- (d) $\lambda_{\min}(T_M)$ *is non-increasing in M.*
- (e) *(Certificate)* $\lambda_{\min}(T_M) > K s_M$ *implies* $m > 0$.

Proof. $x^* T_M x = \frac{1}{2\pi}\int X(\theta)^* F(\theta) X(\theta)\, d\theta$ with $X(\theta) = \sum x_j e^{ij\theta}$; this gives $\lambda_{\min} \geq m$. For (a) take $x_j = \sin\frac{\pi(j+1)}{M+1} e^{-ij\theta_0} u$ and use $u^* F u - m \leq K(2 - 2\cos(\theta - \theta_0))$; the quadratic form of $2 - 2\cos(\theta - \theta_0)$ is the Dirichlet Laplacian conjugated by a diagonal unitary. For (b) use $F - m \geq c(2 - 2\cos(\theta - \theta_0))I$ and the same comparison. For (c), $4(\cos\theta - \cos\theta_0)^2 = |h(e^{i\theta})|^2$ with $h(z) = (z - e^{i\theta_0})(z - e^{-i\theta_0})$, so $T_M[|h|^2] = H^* H$ for the convolution $H = H_1 H_2$ by the two linear factors, each with $\sigma_{\min}^2 = s$. (d) is interlacing (C9); (e) follows from (a). □

Lean core: the eigen-equation of the comparison operator, $-\sin(j\theta) + 2\sin((j+1)\theta) - \sin((j+2)\theta) = (2 - 2\cos\theta)\sin((j+1)\theta)$ with the ghost values $\sin 0 = \sin\pi = 0$, and $2 - 2\cos\theta = 4\sin^2(\theta/2)$ [Lean: `dirichlet_three_term`, `dirichlet_left_ghost`, `dirichlet_right_ghost`, `two_sub_two_cos`]. Numerical: bounds hold for 9 Hermitian symbols and 4 even symbols; the ratio $(\lambda_{\min} - m)/s_M$ tends to $\lambda''(\theta_0)/2$ (1.6576 vs 1.6634 at $M = 320$). In case (c) the true order M^{-2} is [Numerical] only; it is expected from the literature on extreme eigenvalues of Toeplitz matrices [15, 3] (to be checked).

4.6. **Twisted circulants and a zero count (I.7).**

Theorem 4.7. [Closed] *Let S_φ be the twisted shift ($S_\varphi^N = e^{i\varphi}$) and $C_\varphi = \sum_d S_\varphi^d \otimes L(a_d)$ in $\mathrm{Cl}(p,q)$, complex coefficients.*

- (a) *$C_\varphi \cong \bigoplus_k L(P(z_k))$ unitarily, $z_k = e^{i(\varphi + 2\pi k)/N}$.*
- (b) *The singular values are $\sigma_\pm(P(z_k))$, and $\dim\ker C_\varphi = \sum_k \kappa(P(z_k))$ with $\kappa(p) = 2^n$ if $p = 0$, 2^{n-1} if $p \neq 0 = N(p)$, and 0 otherwise.*
- (c) *As φ runs over $[0, 2\pi)$, $\sum_\varphi \dim\ker C_\varphi / 2^{n-1} = \#\{z \in S^1 : T(z) = 0 \neq P(z)\} + 2\#\{z \in S^1 : P(z) = 0\}$; zeros of $T = N(P)$ off the unit circle never produce a kernel, and the count does not depend on N.*
- (d) *$\frac{|T(z_k)|}{\sqrt{2}\|P(z_k)\|} \leq \sigma_- \leq \frac{|T(z_k)|}{\|P(z_k)\|}$ mode by mode.*

Proof. S_φ is unitary with eigenvectors $(z_k^j)_j$; (b) follows from Theorem 4.2; for (c), $(\varphi, k) \mapsto z_k$ is a bijection $[0, 2\pi) \times \mathbb{Z}_N \to S^1$; (d) as in Corollary 4.4. □

Numerical: $\mathrm{Cl}(0,3)$ example with a common zero and an isotropic zero (count 3), $\mathrm{Cl}(1,1)$ example (count 4 for $N = 5, 7, 11, 13$).

4.7. **Perturbation (I.8).** [Closed] For symmetric block circulants, $|\Delta\lambda_{\min}| \leq \max_k(|\Delta P_0| + |\Delta\vec{P}|)(\omega_k) \leq \sum_d(|\Delta a_{d,0}| + |\Delta\vec{a}_d|)$, by Weyl's inequality (C9) mode by mode; the first bound is attained.

4.8. **Persymmetric operators: full spectrum and chiral structure (I.9).**

Theorem 4.8. [Closed] *Let $H_{rc} = L(a_{(r+c) \bmod N})$ in $\mathrm{Cl}(n,0)$ with real a_d, and $P_k = \sum_d a_d \omega^{kd}$.*

- (a) *$H(f_k \otimes u) = f_{-k} \otimes L(P_k)u$. On a pair $\{k, -k\}$, H is the block $\left(\begin{smallmatrix} 0 & L(P_k)^* \\ L(P_k) & 0 \end{smallmatrix}\right)$, with eigenvalues $\pm\sigma_\pm(P_k)$.*
- (b) *On a self-conjugate mode ($k = 0$, and $k = N/2$ for even N) the eigenvalues are $P_0 \pm |\vec{P}|$.*
- (c) *The signature of H is $\sum_{z=\pm 1} 2^{n-1}[\mathrm{sgn}(P_0 + |\vec{P}|) + \mathrm{sgn}(P_0 - |\vec{P}|)](z)$.*
- (d) *For $N \geq 3$, $H \succeq 0$ iff $a_d = \alpha + \beta(-1)^d$ and $P(\pm 1)$ lie in the closed future cone.*
- (e) *With $\Gamma = \sum_{0<k<N/2}(\Pi_k - \Pi_{-k}) \otimes I$: $\Gamma H + H\Gamma = 0$, $\Gamma^2 = 1 - \Pi_{\mathrm{sc}}$, and $H^2 = \bigoplus L(P_k)^* L(P_k)$.*

Proof. (a) $\sum_c L(a_{r+c})\omega^{kc} = \omega^{-kr} L(P_k)$; H is real symmetric and $L(P_{-k}) = L(\bar{P}_k) = L(P_k)^*$. (b) Lemma 4.1. (c)–(d) Pairs contribute equally many positive and negative eigenvalues. (e) H maps V_k to V_{-k}. □

Lean core: the paired block squares to $\mathrm{diag}(BA, AB)$, anticommutes with $\Gamma = \mathrm{diag}(1, -1)$, $\Gamma^2 = 1$, and $Hv = \mu v \Rightarrow H\Gamma v = -\mu\Gamma v$ [Lean: `chiralBlock_sq`, `chiralGrading_anticomm`, `chiralGrading_sq`, `chiral_eigen_reflect`]. Numerical: full spectra for $n \le 4$, $N \le 9$; none of 720 random $N \ge 3$ examples is positive semidefinite.

Remark 4.9. Theorem 4.2 is the spectrum of H^2: the singular-value formula is the spectrum of the square of a Dirac-type operator, and the signature of H is carried by the unpaired (boundary) modes $z = \pm 1$ only. Hankel-type cyclic structure is therefore nearly incompatible with positivity; positivity lives in the circulant and Toeplitz forms (I.5, I.6).

4.9. **Null paravectors in $\mathrm{Cl}(n, n)$ (I.10).** [Closed] Every nonzero real paravector $P \in \mathrm{Cl}(n, n)$ with $N(P) = 0$ has rank $L(P) = 2^{2n-1}$. Proof: $L(P)L(\bar{P}) = 0$ and $L(P) + L(\bar{P}) = 2p_0$; the grade involution gives equal ranks, hence half rank when $p_0 \neq 0$; when $p_0 = 0$, $P^2 = 0$ and some u has $uP + Pu = 1$, and the abstract half-dimension lemma applies [Lean: `twice_finrank_ker_of_square_zero_and_anticommutator`].

5. Layer II: the Laguerre–Clifford tower and single-point criteria

5.1. Jensen identity and the Clifford embedding (II.2, II.3).

Theorem 5.1. [Classical / Closed] *(II.2)* $|f(x + iy)|^2 = \sum_n L_n[f](x) y^{2n}$ *for real entire* f *(C10).* *(II.3)* $L_n = c_n^2 + \sum_{j<n} (-1)^{n+j} \left[\frac{1}{2}(c_j + c_{2n-j})^2 - \frac{1}{2}(c_j - c_{2n-j})^2\right]$, *the norm form of signature* $(n+1, n)$; *hence* $L_n = N(P_n)$ *for a paravector* $P_n \in \mathrm{Cl}(n, n) \cong M_{2^n}(\mathbb{R})$, *and* $L(P_n)L(\bar{P}_n) = L_n \cdot I$.

Proof. The Cauchy coefficient of y^m in $\sum c_j (iy)^j \sum c_k (-iy)^k$ is $i^m \sum_k (-1)^k c_k c_{m-k}$; it vanishes for odd m by the reflection $k \mapsto m - k$, and for $m = 2n$ equals L_n. For II.3 split the sum at its middle term and pair j with $2n - j$; $2uv = \frac{1}{2}[(u + v)^2 - (u - v)^2]$. □

Lean: [Lean: `alternating_cauchy_odd`, `laguerreForm_eq_even_cauchy`, `I_pow_mul_neg_I_pow`, `I_pow_two_mul`, `laguerreForm_signature`] (the Cauchy-product step itself is standard and not formalised). The matrix identity $L(P_n)L(\bar{P}_n) = L_n I$ is checked exactly for $n \le 3$ (`stage2_exact.py`).

5.2. RH and the tower (II.4–II.7, II.9).

Theorem 5.2. *(a)* *(II.4)* [Classical] *RH iff* $N(P_n(x)) = L_n[\Xi](x) \ge 0$ *for all* n *and real* x.

(b) *(II.5)* [Closed-cond] *If* $\Xi(\gamma - i\delta) = 0$ *with* $\delta \neq 0$, *some* $L_n[\Xi](\gamma) < 0$ *with* $n \ge 1$.

(c) *(II.6)* [Closed] $L_n[f]'' + (2n+1)(2n+2)L_{n+1}[f] = 4L_n[f']$; *and* $N(P_2[f]) = \frac{1}{3}N(P_1[f']) - \frac{1}{12}(f''^2 - ff'''')$; *and* $L_1^2 - 2L_0L_2 = -\frac{f^4}{6}(\log f)''''$.

(d) *(II.7)* [Closed-cond] *Unconditionally,* $L_n[\Xi](x) = \Xi(x)^2 e_n(x)$ *and* $p_k(x) = -(\log \Xi)^{(2k)}(x)/(2k-1)!$.

(e) *(II.9)* [Closed-cond] *Under RH,* $L_n^2 \ge (1 + \frac{1}{n})L_{n-1}L_{n+1}$ *(the tower is log-concave).*

Proof. (a) "⇒": each factor of (C1) contributes $|1 \mp z/\gamma|^2 = ((x \mp \gamma)^2 + y^2)/\gamma^2$, a polynomial in y^2 with non-negative coefficients. "⇐" is (b): $0 = |\Xi(\gamma - i\delta)|^2 = \sum L_n(\gamma)\delta^{2n}$, and not all terms can vanish. (c) Compare y^{2n} in $\Delta|f|^2 = 4|f'|^2$; the other two are direct expansions. (d) $\Xi(x + iy)\Xi(x - iy) = \Xi(x)^2 \prod_\rho (1 + y^2 w_\rho(x))$ by (C1), absolutely convergent since $\sum |w_\rho| < \infty$; take logarithms for p_k. (e) Newton (C8) for positive w_ρ, then a limit. □

Numerical (exact rationals): (c) for random polynomials up to degree 12; (d) for random real-rooted and complex-rooted polynomials; on Ξ at 110 digits, $|e_n - L_n/\Xi^2|_{\mathrm{rel}} \le 10^{-108}$ at five points.

5.3. **The second difference of an off-line pair (II.8).**

Theorem 5.3. [CLOSED] *If $f = ((x-\gamma)^2+\delta^2)g$ with g real, then with $b_n = \delta^{2n}L_n[g](\gamma)$ (and $b_{-1} = b_{-2} = 0$),*

$$\delta^{2n}L_n[f](\gamma) = \delta^4(b_n - 2b_{n-1} + b_{n-2}).$$

Proof. $|f(\gamma+iy)|^2 = (\delta^2-y^2)^2|g(\gamma+iy)|^2$; compare coefficients of $Y = y^2$. □

Lean: the coefficient step for $(\delta^2 - Y)^2 G(Y)$ `[Lean: coeff_offline_factor, coeff_offline_factor_zero]`. Exact check for 20 random g.

5.4. **Single-point matrix criteria (II.1, II.11).**

Theorem 5.4. *Fix a real x with $\Xi(x) \neq 0$. The following are equivalent.*

(*a*) *RH.*
(*b*) [CLOSED-COND] *The Hankel matrices $(p_{j+k+1}(x))_{j,k<N}$ are positive semidefinite for all N.*
(*c*) [CLOSED-COND] *The Toeplitz matrix $(L_{j-k}[\Xi](x))_{j,k}$ is totally nonnegative.*

For $x = 0$, (b) is (II.1): with $\mu_n = \sum_{\gamma>0}\gamma^{-2(n+1)}$, RH iff $(\mu_{j+k}) \succeq 0$ for all N.

Proof. (a)⇒(b): $p_{j+k+1} = \sum w\, w^{j+k}$ with $w > 0$. (b)⇒(a): $|w_\rho| \leq R$ and $\sum|w_\rho| < \infty$; by (C4) there is a unique positive measure ν on $[-R,R]$ with these moments, and $\int \frac{d\nu(s)}{z-s} = \sum_\rho \frac{w_\rho}{z-w_\rho}$ off $[-R,R]$. The right side has poles at the w_ρ with residues (multiplicity)$\cdot w_\rho \neq 0$; analyticity of the left side off the real axis forces all w_ρ real, and positivity of ν forces the total mass at each point to be positive, so $w_\rho > 0$ and t_ρ is real. (A zero with $\operatorname{Re} t = x$ gives the negative atom $-1/\delta^2$ and is excluded too.) For $x = 0$ use (C2). (a)⇒(c): each factor $1 + wY$ gives a totally nonnegative bidiagonal Toeplitz matrix; products preserve total nonnegativity (Cauchy–Binet). (c)⇒(a): Edrei (C5). □

Remark 5.5. [CLOSED] The single-point tower condition $L_n(x) \geq 0$ for all n is *only necessary*: in an exact model with real zeros $-6, -3, -1, 2, 4, 7$ and the pair $\frac{5}{2} \pm \frac{i}{3}$, all $L_n(x_0) \geq 0$ at $x_0 \in \{-\frac{9}{2}, 0, \frac{1}{2}\}$, while (b) fails at $N = 7, 5, 4$ and (c) fails at minors of size $\geq 16, 12$ (`phase2_II11*.py`).

Numerical: $\mu_0 = 0.0231049931154\ldots$; Hankel pivots at $x = 0$ positive for $N \leq 8$; Hankel pivots at $x \in \{0, 5, 17.5, 35, 62.3\}$ positive for $N \leq 4$.

5.5. **The Li–Schoenberg Toeplitz form (II.10).**

Theorem 5.6. [CLOSED-COND] *RH iff $(e^{-t\lambda_{|j-k|}})_{j,k} \succeq 0$ for all $t > 0$ and all sizes.*

Proof. Under RH each conjugate pair contributes $e^{-2t(1-\cos m\theta)} = e^{-2t}\sum_k I_k(2t)e^{ikm\theta}$ with $I_k > 0$, a positive measure on the circle; Schur products and limits preserve positivity. Conversely the 2×2 minors already give $\lambda_n \geq 0$; apply (C6). □

Remark 5.7. The logical strength is exhausted at size 2 (Li's criterion). The value of the form is structural: it presents RH as a Toeplitz matrix whose symbol is a positive measure, the setting of I.5–I.6.

Numerical: $\lambda_n = \sum_j j\binom{n}{j}a_j$ from the Taylor coefficients of $\log\xi$ at 1; λ_1 agrees with $1 + \gamma/2 - \frac{1}{2}\log 4\pi$ to 28 digits, $\lambda_2 = 0.0923457352$; $\lambda_n > 0$ for $n \leq 80$; all 81×81 matrices for $t \in \{0.01, 0.1, 1, 10\}$ positive definite in 110-digit elimination.

5.6. **The depth theorem (II.12).**

Theorem 5.8. [CLOSED-COND] *Let $\rho_0 = \frac{1}{2} + \delta + i\gamma_0$ ($\delta > 0$) be a zero, and suppose the other off-line zeros are finitely many and height-separated: $|\gamma' - \gamma_0| \geq \sqrt{3}\,\delta'$. Let q be the number of quadratic factors (the mirror pair at $-\gamma_0$ and the other off-line pairs) and $m = \sum_k \delta^2/((\gamma_0 - t_k)^2 + \delta^2)$ over the real zeros. Then $L_{n^*}[\Xi](\gamma_0) < 0$ for some $n^* \leq \lfloor m \rfloor + 3 + 2q$; for a single quadruple $n^* \leq \lfloor m \rfloor + 5$. Moreover $m = O(\log \gamma_0)$, so $n^* = O(\log \gamma_0)$.*

Proof. Write $\Xi = ((x - \gamma_0)^2 + \delta^2)G$ and $b_n = \delta^{2n} L_n[G](\gamma_0)$. (*A*) The real-zero part of $|G(\gamma_0 + iy)|^2$ is a product of factors $1 + \delta^2 w_k Y$; each other pair contributes $|(d + iy)^2 + \delta'^2|^2 = (d^2 + \delta'^2)^2 + 2(d^2 - \delta'^2)y^2 + y^4$, whose coefficients are positive and log-concave iff $d^2 \geq 3\delta'^2$. Convolutions of positive log-concave sequences are log-concave (2×2 Cauchy–Binet), and limits preserve this; so b is positive and log-concave. (*B*) By Theorem 5.3, $\delta^{2n} L_n(\gamma_0) = \delta^4 \Delta^2 b_n$, and $\Delta^2 b_{M+1} < 0$ at the last maximum M of b. (*C*) Convolution with a positive sequence of length three moves the start of strict decrease by at most two places, so $M \leq M_a + 2q$ where M_a is the last mode of the real-zero part. (*D*) By (C7) applied to finite truncations and a limit, $M_a \leq \lfloor m \rfloor + 2$ (strict decrease of ratios by Newton's inequalities (C8) excludes flat tops). (*E*) Splitting the real zeros into unit intervals around γ_0 and using (C3), $m \leq C \log \gamma_0 + C'$ uniformly in $\delta < \frac{1}{2}$. □

Lean: steps (A) expansion and criterion, (B) last-mode lemma, (C) shift lemma `[Lean: offline_factor_expansion, offline_factor_logconcave_iff, second_difference_neg_at_last_mode, convolution_strictly_decreasing]`. Exact check of the whole chain on 40 polynomial models (finite form $n^* \leq \lfloor m \rfloor + 2 + 2q$). With the actual zero configuration below height 100, $m < 1$ for $\gamma_0 \leq 80$, $\delta < \frac{1}{2}$, so $n^* \leq 5$; the average-density heuristic $m \approx \frac{\delta}{2} \log \frac{\gamma_0}{2\pi}$ gives $n^* \leq 12$ at $T = 10^{13}$.

5.7. **The index theorem (II.15).**

Theorem 5.9. [CLOSED-COND] *For real x with $\Xi(x) \neq 0$ let $n_-(N)$ be the number of negative eigenvalues of $H_N(x) = (p_{j+k+1}(x))_{j,k<N}$. Then $n_-(N)$ is non-decreasing and*

$$\operatorname{ind}(x) := \lim_{N \to \infty} n_-(N) = \#\{\text{distinct non-real atom pairs}\} + \#\{\text{distinct negative atoms}\}.$$

For generic x this is $2\times$(number of off-line quadruples); at $x = 0$ it is the number of quadruples (the atoms of t and $-t$ coincide). In particular RH iff $\operatorname{ind}(x) = 0$.

Proof. $H_N = \sum_\rho w_\rho v_\rho v_\rho^{\mathsf{T}}$, $v_\rho = (w_\rho^j)_{j<N}$, absolutely convergent. Positive atoms give a positive semidefinite part; a conjugate pair $(c, \bar{c})$ gives $2\,\mathrm{Re}(cvv^{\mathsf{T}})$, whose 2×2 form in the basis $(\mathrm{Re}\, v, \mathrm{Im}\, v)$ has negative determinant, hence one positive and one negative eigenvalue; a negative real atom gives one negative direction. Monotonicity is interlacing; the upper bound is subadditivity of n_-. For the lower bound, for finitely many chosen pairs take polynomials $q(w) = w^M r(w)$ with r vanishing on the finitely many atoms of modulus $\geq \varepsilon$ outside the chosen ones and prescribed at the chosen ones; the remaining atoms contribute $O((\varepsilon/|c|)^{2M})$, so a negative-definite subspace of the required dimension exists for large M. □

Lean core: the conjugate-pair block $\left(\begin{smallmatrix} a & -b \\ -b & -a \end{smallmatrix}\right)$ is traceless with determinant $-(a^2 + b^2) < 0$ `[Lean: conjPairBlock_det, conjPairBlock_det_neg, conjPairBlock_trace]`. Numerical: exact inertia (characteristic polynomial and Descartes' rule) in 16 general and 6 even models; in a realistic model replacing the zeros $49.77, 52.97$ of Ξ by $\pm 51.37 \pm 0.1i$, n_- reaches 2 at x near γ_0, while the unmodified Ξ gives $n_- = 0$ everywhere.

Remark 5.10. In the language of Layer I, on-line zeros are self-conjugate modes and off-line pairs are chiral pairs: RH says that the Hankel form of the atoms has no chiral pair. The finite case

is Hermite's classical theorem on Hankel forms of power sums [11]; the entire-function version belongs to the theory of kernels with finitely many negative squares [16, 10], where the statement for Ξ may already be known.

6. Observations and conjectures

Observation 6.1. [Numerical](II.16, beam width) In the realistic model above ($\delta = 0.1$), the tower detects the hidden zero only for $|x - \gamma_0| \leq 0.05$, while the Hankel index detects it at $N = 1, 1, 2, 2, 3, 3, 5, 8$ for $x - \gamma_0 = 0, 0.05, 0.2, 0.5, 1, 2, 3, 5$.

Conjecture 6.2. [Conjecture](II.13, resolution limit) Any criterion built from the derivatives of order at most $2n$ at one point misses off-line pairs with $\delta/s \gg n$, where s is the local zero spacing.

Observation 6.3. [Numerical] On $[0, 100]$ the normalised margins $\kappa_n = N(P_n)/\|P_n\|^2$ of Ξ have minima 0.206, 0.103, 0.063 for $n = 1, 2, 3$, located in wide zero gaps; the Newton ratios of Theorem 5.2(e) are tightest at the symmetric point $t = 0$ (1.036, 1.034).

7. Lean correspondence

Item	Lean declaration(s)	Module
I.1	`sum_paravectorKernel_finrank_eq_corrected_formula`	`NullityFormula` (existing core)
I.2, I.3	`eigenvalue_sq_of_quadratic_relation`, `paravectorMul_quadratic_relation`, `paravectorMul_eigenvalue`, `euclidean_paravector_eigenvalue`	`Positivity/QuadraticRelation`
I.3	`commutator_of_scalar_relations`, `clifford_commutator_relations`	`Positivity/TwoVectorAlgebra`
I.5	`euclidean_paravector_eigenvalue_nonneg`; DFT reduction	`QuadraticRelation`; `FourierReduction`
I.6	`dirichlet_three_term`, `dirichlet_left_ghost`, `dirichlet_right_ghost`, `two_sub_two_cos`	`Positivity/DirichletLaplacian`
I.9	`chiralBlock_sq`, `chiralGrading_anticomm`, `chiralGrading_sq`, `chiral_eigen_reflect`	`Positivity/ChiralBlock`
I.10	`twice_finrank_ker_of_square_zero_and_anticommutator`	`KernelRank` (existing core)
II.2, II.3	`alternating_cauchy_odd`, `laguerreForm_eq_even_cauchy`, `laguerreForm_signature`	`Laguerre/JensenCoefficients`
II.8	`coeff_offline_factor`, `coeff_offline_factor_zero`	`Laguerre/SecondDifference`
II.12	`offline_factor_logconcave_iff`, `second_difference_neg_at_last_mode`, `convolution_strictly_decreasing`	`Laguerre/SecondDifference`
II.15	`conjPairBlock_det_neg`, `conjPairBlock_trace`	`Positivity/ChiralBlock`

Not formalised: spectral multiplicities, the adjoint identity of Theorem 4.2, the Toeplitz integral identity, the analytic statements resting on (C1)–(C10) (Mathlib contains the definition of RH and the zeta function, but not the Hadamard product of Ξ or zero counting [13]).

8. Literature status (preliminary)

Item	Assessment	Pointers
II.2, II.4	classical	Jensen; Patrick; Csordas–Varga [5, 4]
II.3 Clifford form	new framing, elementary	—
II.6, II.9	elementary; possibly known	Dimitrov; Cardon [4]
II.7 product form	probably known	Jensen-type product
II.8 second difference	not found yet	—
II.1, II.11(b)	Hermite–Sylvester type; ζ version to be checked	[11, 19]
II.11(c)	corollary of Edrei	[7, 18]
II.10	elementary combination	[12, 2]
II.12 depth	not found yet	Jensen–Pólya literature to be checked [8]
II.15 index	general theory probably known	[16, 10]
I.6 case (c) order	expected known	[15, 3]

9. Numerical verification

All scripts are in `verification/` and run with Python 3 and NumPy (SciPy for I.6); no other package is needed. Ξ is evaluated from Riemann's integral $\Xi(t) = 2\int_0^\infty \Phi(u)\cos(tu)\,du$ by the trapezoid rule ($h = 0.01$, $u \le 3$) in 110-digit decimal arithmetic; the checks $\Xi(0) = 0.497120778188314\ldots$, $\xi(1) = \frac{1}{2}$, and $|\Xi| \approx 10^{-50}$ at the first 29 zeros (given to 50 digits) fix the accuracy.

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